

NYC Airbnb Listings — Reproducible Data Workflow

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Overview

This module implements a complete, reproducible data workflow using the NYC Airbnb Listings dataset. The notebook loads the dataset, applies structured cleaning functions, performs exploratory data analysis, and produces multiple labeled visualizations. The workflow emphasizes reproducibility, modular function design, and clear documentation—forming a foundation for later machine learning, deep learning, generative AI, and agentic systems.

Dataset Description

The dataset contains ~**48,000 Airbnb listings** across New York City, with **16 columns** including price, room type, neighbourhood group, availability, and review metrics.

Key variables used in the workflow include:

- **price** (numeric) — nightly rate
- **neighbourhood_group** (categorical) — borough-level grouping
- **room_type** (categorical) — property type
- **num_of_reviews** (numeric) — activity indicator
- **minimum_nights** (numeric) - minimum number of nights a guest must book to stay at that listing
- **availability_365** (numeric) - number of days available during the year

The dataset is large enough for robust descriptive analysis and contains meaningful categorical and numeric variables suitable for grouping, visualization, and statistical exploration.

Workflow Description

The workflow proceeds in five stages.

- **Ingestion.** The CSV is loaded with Pandas and the first rows are displayed to confirm the data arrived correctly.
- **Cleaning.** Missing values, outliers, and non-rentable listings are addressed using two reusable cleaning functions with docstrings.
- **Exploratory analysis.** A correlation block quantifies how much variance each variable explains on its own, followed by grouped descriptive statistics by borough and property type.
- **Visualizations.** Three figures summarize the univariate distributions and the bivariate dependence of price on location and property type.
- **Summary.** The findings, assumptions, limitations, and surprising patterns are documented.

Key Decisions and Assumptions

Cleaning choices: *First*, price is clipped at the 0.5% and 99.5% quantiles to remove extreme outliers that would distort correlation values and obscure the boxplots. *Second*, shared room listings are dropped so the analysis focuses on private room and whole-property nightly rates.

These choices are not free of consequence. Missing data are frequently **not** missing at random, and common imputation or deletion approaches can shift results and amplify bias (Little & Rubin, 2002). The 1% price clip discards roughly 1% of listings and truncates the heavy upper tail, so summary statistics no longer include the most expensive properties. Removing shared rooms narrows the population to private room and whole-property rentals, so the findings describe that segment rather than all of NYC Airbnb.

Exploratory focus: The EDA block was designed to answer one question — "how much of price does each feature explain on its own?" — using a common yardstick. For the numeric features, Spearman's ρ measures monotonic association; for the categorical features, the correlation ratio η measures how well the grouping predicts price, and η^2 is exactly the ANOVA effect size (the proportion of variance explained).

What each plot was designed to show: Figure 1 summarizes the univariate distributions of price and the two key categorical variables. Figure 2 shows the median nightly price by borough to reveal the location gradient. Figure 3 shows price by property type within each borough to demonstrate that property type raises price independently of location.

Summary & Interpretation

How much does each feature explain on its own? Squaring each statistic puts them on a common scale — the proportion of price variance explained by that variable alone:

Numeric feature	Spearman ρ	Variance explained (ρ^2)
minimum_nights	+0.098	0.96%
availability_365	+0.099	0.98%
number_of_reviews	-0.054	0.30%

Categorical feature	η	Variance explained (η^2)
neighbourhood_group	0.088	0.77%
room_type	0.219	4.8%

Every feature except room_type explains under 1% of price variance on its own. room_type is the lone outlier at ~5%: entire homes/apt consistently outprice private rooms.

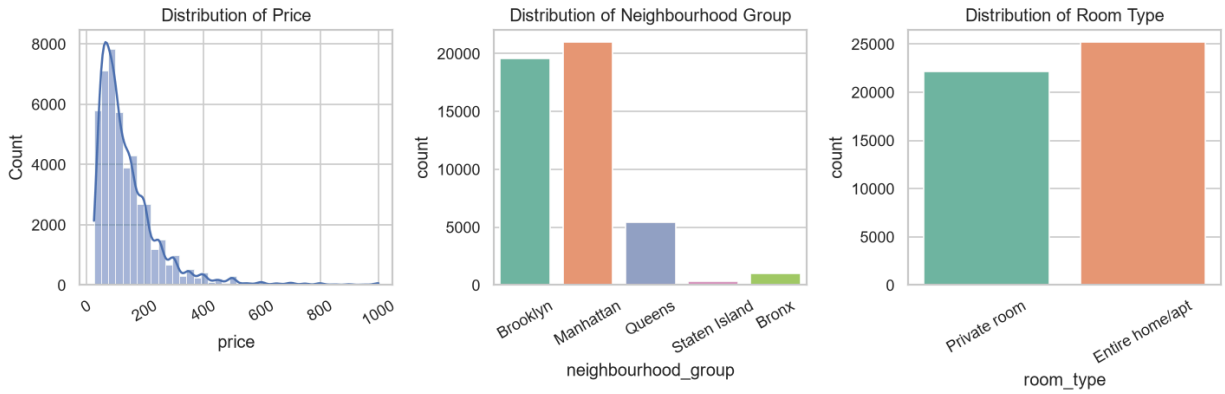


Fig 1. Record Distribution by Price, Neighbourhood Group and Room Type

Location gradient (Figure 2–3):

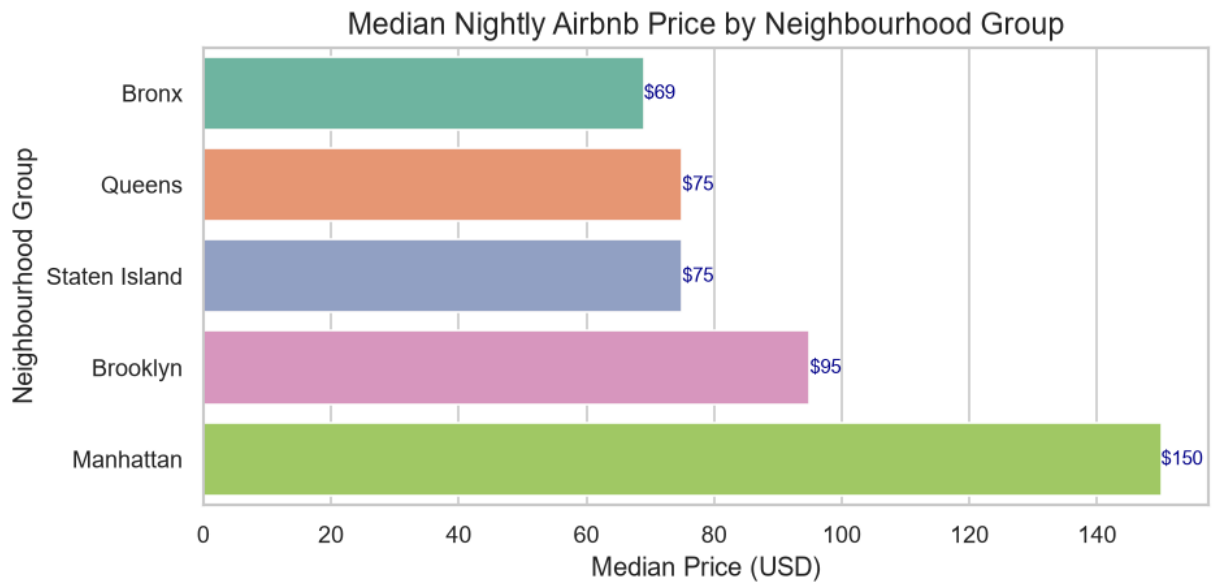


Fig 2. Median Nightly Price by Neighbourhood Group

By neighbourhood, median nightly price falls in an ordered gradient — Manhattan (\$150) > Brooklyn (\$95) > Queens (\$75) $\approx$ Staten Island (\$75) > Bronx (\$69) — with Manhattan's mean ($\approx$ \$181) nearly double the Bronx's ($\approx$ \$88).

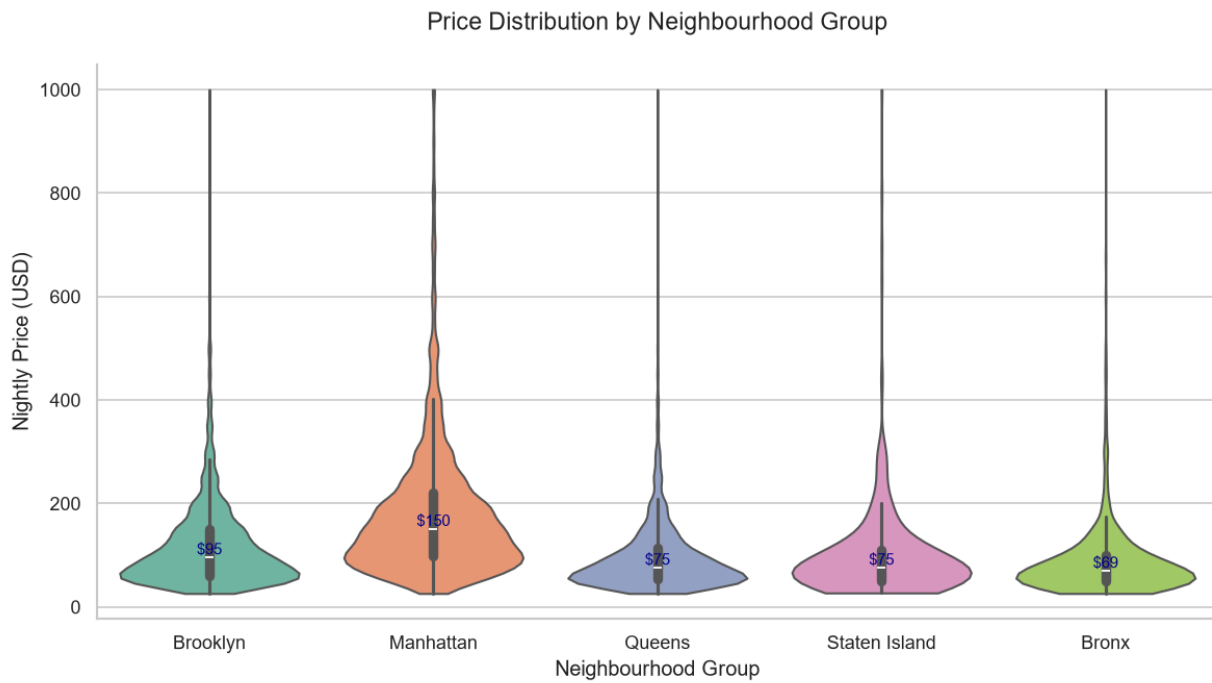


Fig 3. Price Distribution by Neighbourhood Group

Every borough's price distribution is strongly right-skewed, confirming the long upper tail seen in the histogram. Manhattan and Brooklyn show the widest price distributions (supporting a range of listings from budget to luxury), whereas the Queens, Bronx and Staten Island are tightly clustered around lower prices.

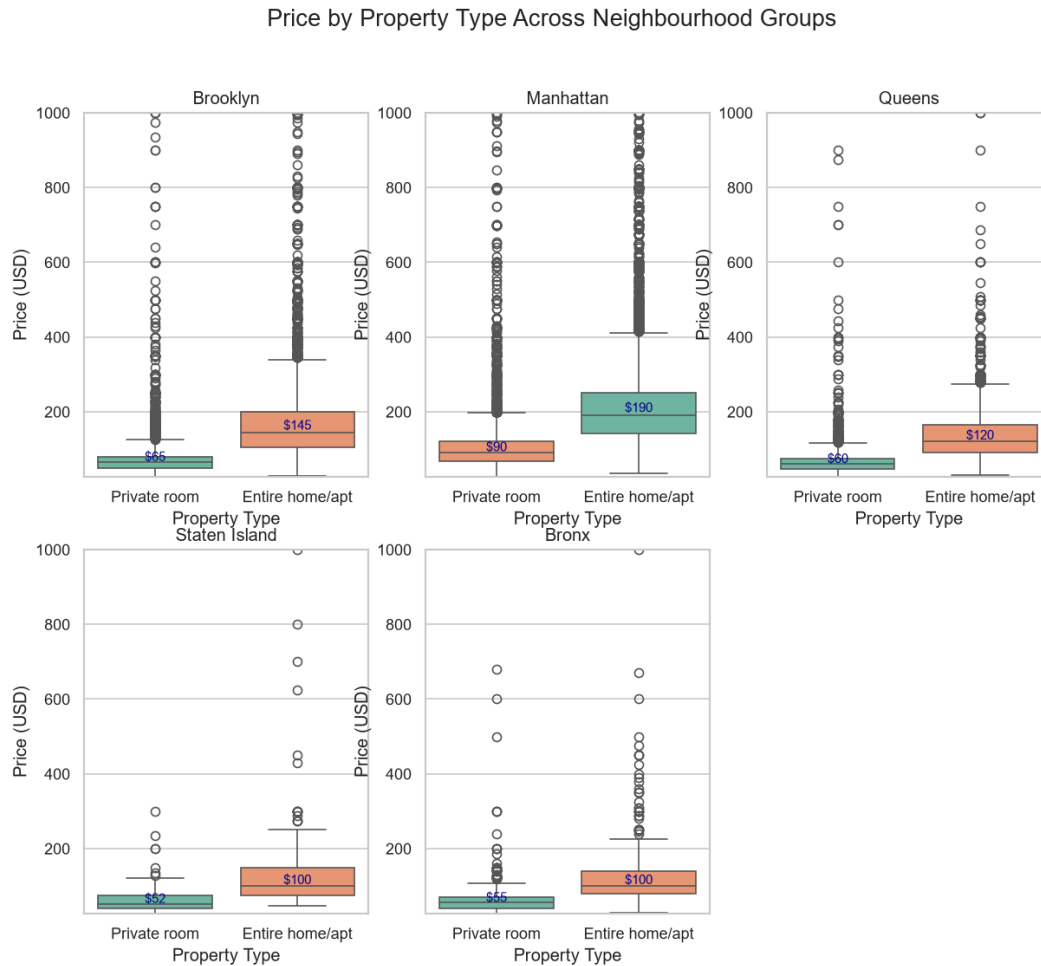


Fig 4. Price by Room Property Type across Neighbourhood Groups

Property type (Figure 4): Within every borough, "Entire home/apt" listings command a higher median than "Private room" listings, and this stratification holds across all five boroughs. Together, Figures 2–4 show that **both location (neighbourhood) and property type independently raise price**, with Manhattan's location effect dominating.

Why is the neighbourhood's η^2 so small (0.77%) despite a visible Manhattan–Bronx gap? Because price is heavily right-skewed, the large within-group spread and the extreme tail inflate the denominator of η , shrinking it. The median gap is real and clearly visible in Figures 2–3; the correlation ratio is computed on the mean, which skews badly under this distribution.

For such heavily skewed data the median is a more robust measure of central tendency than the mean (Wilcox, 2012), which is why the grouped statistics and plots report the median.

Takeaway: Both the correlation results and the grouped statistics agree: price is predominantly a function of property type, to a lesser extent location, and essentially independent of minimum-nights, review frequency, and availability in this univariate analysis. Yet even the strongest single feature explains less than 5% of price variance — price is driven more by the wide spread within each category than by these features alone, which is exactly why a predictive model would need more predictors than any of these univariate measures.

Responsible Practice (Bias and Data Quality)

Several cleaning and handling decisions could introduce bias or misleading results. **First**, clipping the 0.5%/99.5% quantiles discards ~1% of listings and truncates the heavy upper tail, so the reported medians no longer reflect the most expensive listings. **Second**, dropping shared room listings narrows the population to whole-property rentals, so results describe that segment rather than all NYC Airbnb. **Third**, the Bronx and Staten Island have the fewest listings, making their medians more sensitive to outliers and less stable.

Reproducibility

Anyone can rerun this work from the cloned repository. The **requirements.txt** file, generated with **pip freeze**, pins every dependency so the environment can be recreated exactly (**pip install -r requirements.txt**). The code is organized into modular, reusable cleaning and EDA functions, each with an informative docstring, and the entire workflow is tracked with Git: development work is carried out on a separate feature branch and integrated back into main through commits, which supports auditability and iteration (Wright, 2023).

Conclusion

This module establishes a clean, reproducible data workflow that meets all rubric requirements. It provides a strong foundation for future modules involving machine learning, deep learning, generative AI, and agentic systems by ensuring the dataset is well-understood, well-cleaned, and well-documented.

References

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